

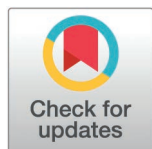
RESEARCH ARTICLE

The mediating roles of personal mastery and health-promoting behaviors in the relationship between self-regulatory fatigue and quality of life among patients with type 2 diabetes mellitus

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Data availability statement: All data supporting the findings of this study are fully available without restriction. The anonymized dataset used for analysis has been uploaded as a [Supporting Information](#) file (S2 Dataset).

Abstract

Background

Quality of life (QoL) is a pivotal prognostic indicator for patients with Type 2 Diabetes Mellitus (T2DM). While the impact of various factors on QoL is recognized, the complex interplay between psychological and behavioral determinants remains underexplored. Understanding these pathways is essential for developing effective interventions.

Objective

This study aimed to investigate the status of QoL and its associated factors among T2DM patients, and to examine the mediating effects of personal mastery (PM) and health-promoting behaviors (HPB) in the relationship between self-regulatory fatigue (SRF) and QoL by using structural equation modeling.

Methods

The study adhered to the STROBE guidelines for cross-sectional reporting. From July 2025 to February 2026, 432 T2DM patients were recruited from three tertiary grade-A hospitals across two cities in China by using convenience sampling. Data were collected using the Demographic Characteristics Questionnaire, Self-Regulatory Fatigue Scale, Personal Mastery Scale, Diabetes Health Promotion Scale, and Diabetes-Specific Quality of Life Scale. Pearson's correlation, multiple linear regression, and structural equation modeling were employed for data analysis.

xlsx). This file includes all raw, de-identified data points necessary to replicate the analyses. Analytical methods are fully described in the manuscript. No ethical or legal restrictions prevent data sharing. For any additional inquiries, the corresponding authors can be contacted, but the data are already publicly accessible via the journal's supplementary materials.

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Competing interests: The all authors declare that there is no conflict of interest.

The mediating effects were tested using the bias-corrected (BC) bootstrap method with 5,000 resamples.

Results

The total scores for SRF, PM, HPB, and QoL were 38.63 ± 12.84 , 26.12 ± 7.62 , 97.69 ± 20.87 , and 69.41 ± 20.89 , respectively. Correlation analysis indicated that QoL was significantly positively correlated with SRF ($r=0.581$, $p<0.01$), and negatively correlated with PM ($r=-0.557$, $p<0.01$) and HPB ($r=-0.613$, $p<0.01$). Multiple linear regression analysis revealed that age, educational level, medical insurance payment method, smoking status, glycemic control status, number of comorbidities, self-care ability, self-regulatory fatigue, personal mastery, and health-promoting behaviors were independent predictors of QoL. Furthermore, the structural equation modeling demonstrated a significant total indirect effect of self-regulatory fatigue on QoL. The mediating effect of personal mastery accounted for 30.2% of the total indirect effect, while that of health-promoting behaviors explained 60.2%. The chain mediating pathway contributed 9.5% of the total indirect effect.

Conclusion

This study suggests that SRF is statistically associated with directly and indirectly poorer QoL in T2DM patients, primarily by being associated with lower PM and fewer HPB. These findings highlight the importance of addressing psychological and behavioral mechanisms in diabetes care. From a nursing perspective, interventions targeting SRF, strengthening PM, and promoting HPB may be beneficial.

1. Introduction

Diabetes mellitus encompasses a group of metabolic disorders characterized by chronic hyperglycemia resulting from defects in insulin secretion, insulin action, or both. The condition includes Type 1 Diabetes, Type 2 Diabetes (T2DM), other specific types, and gestational diabetes [1]. Chronic hyperglycemia in diabetes is associated with long-term damage, dysfunction, and failure of various organs, particularly the eyes, kidneys, nerves, heart, and blood vessels, and may also lead to acute metabolic complications such as diabetic ketoacidosis. The disease is typically progressive, requires lifelong management, has low cure rates, and is associated with high risks of disability and premature mortality [2].

China currently has the largest diabetic population worldwide, with T2DM representing the predominant form. According to recent epidemiological data, approximately 148 million adults in China were living with diabetes in 2024, corresponding to a prevalence of 11.2%. The disease exhibits a gender disparity, with higher prevalence observed in men than in women [3]. The economic implications are substantial; global direct health expenditure on diabetes was estimated at USD 825 billion annually, and China's share is projected to reach USD 170 billion

by 2025, positioning China as the country most economically burdened by diabetes globally [4]. Moreover, as highlighted in the Guidelines for the Prevention and Treatment of Diabetes in China (2024 Edition) [5], the disease burden is escalating exponentially, imposing mounting economic pressure on families, healthcare systems, and society at large.

Quality of life (QoL) refers to an individual's subjective perception of their life situation, framed within the context of their cultural and value systems [6]. It is a multidimensional construct encompassing physical health, psychological state, level of independence, social relationships, and personal beliefs. Evidence suggests that enhancing QoL can facilitate patients' reintegration into normal life and society, and promote a more proactive attitude toward disease self-management [7]. Higher QoL has been associated with improved emotional well-being, greater adherence to treatment regimens, reduced risk of diabetes-related complications, and delayed disease progression [8].

The construct of self-regulatory fatigue (SRF) describes the depletion of self-regulatory resources following repeated or sustained efforts to manage stress or difficult situations, based on the premise that such resources are limited [9]. In the context of T2DM, where patients must continually regulate their behavior to maintain glycemic control, SRF may impair their capacity to sustain health-promoting routines, thereby contributing to poor clinical outcomes. SRF can exacerbate diabetes-related impairments across physiological, cognitive, behavioral, and social domains, potentially compounding fatigue and diminishing QoL.

Personal mastery (PM) reflects the extent to which individuals perceive themselves as having control over events and outcomes that matter in their lives [10]. As a core psychological resource, PM enables individuals to cope more effectively with stress and chronic illness. In the context of T2DM, higher levels of personal mastery have been linked to greater engagement in standardized self-care behaviors, improved glycemic stability, reduced risk of complications, enhanced physical comfort, and, consequently, better QoL [11].

Health-promoting behaviors (HPB) are proactive measures individuals undertake to maintain or improve their health [12]. These include adherence to prescribed medications, dietary modification, regular physical activity, stress management, and routine health monitoring. In T2DM patients, consistent engagement in HPB contributes directly to metabolic control, lowers the likelihood of complications, and enhances both physiological and psychological well-being.

While previous studies have independently established the relationships between SRF and QoL, PM and self-management, or HPB and metabolic control, the sequential, causal-chain mechanism linking all four constructs remains largely untested. Specifically, no research has investigated whether PM acts as a "meta-resource" that, once depleted by SRF, subsequently compromises HPB, leading to reduced QoL. Drawing upon the Conservation of Resources (COR) theory and the Theory of Planned Behavior, we propose a hierarchical depletion model. This model posits that SRF first erodes a higher-order psychological resource (PM), which in turn leads to a decline in specific behavioral actions (HPB), ultimately worsening QoL. To address this gap, the present study employed a cross-sectional design to: (1) investigate the status of QoL and its associated factors among patients with T2DM, and (2) examine the mediating effects of PM and HPB in the relationship between SRF and QoL by using structural equation modeling.

Based on the literature review, stress coping theory and Roy's adaptation model, the conceptual framework of self-regulatory fatigue, personal mastery, health-promoting behaviors, and quality of life of this study was constructed, as shown in Fig 1.

2. Objectives

This study aimed to (1) investigate the status of QoL and its associated factors among patients with T2DM; (2) to examine the mediating effects of personal mastery and health-promoting behaviors in the relationship between self-regulatory fatigue and QoL by using structural equation modeling.

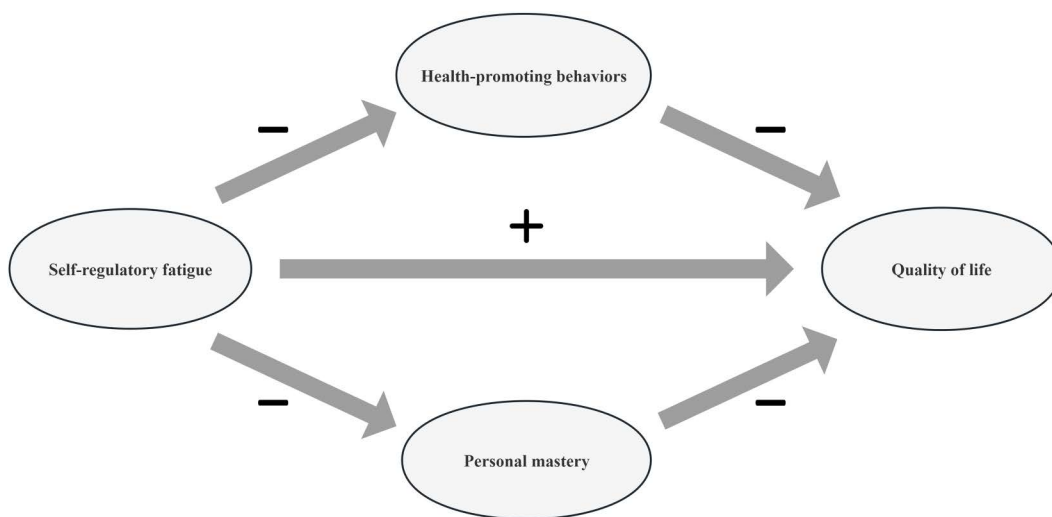


Fig 1. The conceptual framework of self-regulatory fatigue, personal mastery, health-promoting behaviors, and quality of life of the study.

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3. Methods

3.1. Study design and setting

This study employed a descriptive, cross-sectional, multicenter design and was conducted in China. The quality reporting of the study adhered to the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines for cross-sectional studies. Data were collected at a single time point from July 2025 to February 2026. The design does not include a control group, as it is a descriptive and correlational study examining associations among psychological and behavioral variables in a cohort of T2DM patients.

3.2. Participants and sample

3.2.1 Sampling procedure and recruitment. A convenience sampling strategy was employed. From July 2025 to February 2026, trained research nurses approached potential participants during their routine outpatient visits or hospitalizations at the endocrinology departments of three tertiary grade-A hospitals in Tianjin and Hengyang, China. The head nurses of each department generated a preliminary list of patients who met the inclusion criteria. Research nurses then sequentially invited these patients in a private setting, explained the study's aims, assured confidentiality, and obtained written informed consent. Data collection took place in a quiet room adjacent to the clinic to minimize distractions and ensure privacy.

The inclusion criteria were as follows: (1) confirmed diagnosis of T2DM based on American Diabetes Association (ADA) criteria [13]; (2) age ≥ 18 years; (3) disease duration ≥ 6 months; (4) ability to read and complete the questionnaires independently or with researcher assistance; and (5) provision of written informed consent. Exclusion criteria: (1) presence of severe diabetes-related complications (e.g., dialysis-dependent nephropathy, proliferative retinopathy, or history of lower extremity amputation) or life-threatening comorbidities (e.g., end-stage cancer, severe heart failure) that could interfere with survey completion or quality of life; (2) documented psychiatric disorders or significant cognitive impairment affecting capacity to provide consent or complete the questionnaire; or (3) current participation in other interventional clinical trials.

G*Power 3.1.9.7 software [14] was used to calculate the minimum required sample size for F-tests in a linear multiple regression fixed model (R^2 deviation from zero). Parameters were: effect size $f^2 = 0.15$ (medium), α err prob = 0.05, power ($1 - \beta$ err prob) = 0.95. The calculated minimum sample size was 107. To account for the complexity of structural equation

modeling (SEM) and potential attrition, we recruited 432 participants from three tertiary grade-A hospitals. A post-hoc power analysis using the achieved sample size ($n=432$), $R^2=0.15$, and $\alpha=0.05$ yielded a power ($1 - \beta$) of 1.00, indicating that our sample was more than sufficient to detect the hypothesized effects. No control group was included because this was a cross-sectional, descriptive study of a single cohort of T2DM patients. After eliminating 19 questionnaires with obviously contradictory responses (e.g., logically inconsistent answers or identical responses across all items), we obtained 432 completely valid questionnaires with no item-level missing data. Specifically, all 432 included cases had complete responses on every item of the Demographic Characteristics Questionnaire, Self-Regulatory Fatigue Scale, Personal Mastery Scale, Diabetes Health Promotion Scale, and Diabetes-Specific Quality of Life Scale. Therefore, no statistical imputation method (e.g., FIML or multiple imputation) was required. The effective recovery rate was 95.8%.

3.3. Measurements

3.3.1 Demographic characteristics questionnaire. The Demographic Characteristics Questionnaire was rigorously developed and validated by the research team to collect a comprehensive set of participant information. It included variables such as gender, age, body mass index (BMI), educational level, marital status, living conditions, residence, employment status, per capita monthly income, medical insurance payment method, drinking status, smoking status, disease duration, number of comorbidities, treatment modality, glycemic control status, whether combined with other chronic diseases, and self-care ability.

3.3.2. Self-Regulatory Fatigue Scale (SRFS). The Chinese version [15] of the scale originally developed by Nes et al. [16] was employed in this study. This 16-item instrument assesses fatigue across three dimensions—cognitive, emotional, and behavioral control—using a 5-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). The total possible scores range from 16 to 80, with higher scores reflecting greater levels of fatigue. In the present study, the scale demonstrated satisfactory internal consistency, with a Cronbach's α coefficient of 0.857.

3.3.3. Personal Mastery Scale (PMS). The Personal Mastery Scale (PMS) was used to assess participants' perceived control over their lives. The original scale was developed by Pearlin [17]. We utilized the Chinese version of the scale, which was adapted and validated for the Chinese context by Yu et al. [18]. This unidimensional scale comprises seven items, each rated on a 5-point Likert scale ranging from 1 (not at all consistent) to 5 (extremely consistent). Total scores range from 7 to 35, with higher scores reflecting a greater sense of personal mastery. In the present study, the scale demonstrated satisfactory internal consistency, with a Cronbach's α coefficient of 0.886.

3.3.4. Diabetes Health Promotion Scale (DHPS). The Chinese version of the scale [19], adapted from the original work by Chen et al. [20], was utilized in this study. This 28-item instrument assesses health-related behaviors across six distinct dimensions: exercise, stress management, health responsibility, risk avoidance, healthy diet, and life appreciation. Each item is rated on a 5-point Likert scale ranging from 1 (never) to 5 (always), based on the frequency of engaging in each behavior. The total possible scores range from 28 to 140, with higher scores reflecting more positive and frequent engagement in health-promoting behaviors. The scale demonstrated excellent internal consistency in the current sample, with a Cronbach's α coefficient of 0.915.

3.3.5. Diabetes-Specific Quality of Life Scale (DS-QoLS). Developed by Fang et al. [21], this 27-item instrument evaluates quality of life across four multidimensional domains: physical function, psychological/spiritual status, social relations, and treatment. Each item is rated on a 5-point Likert scale reflecting severity, ranging from 1 (none or no problem) to 5 (most severe or extreme difficulty). Total possible scores range from 27 to 135, with higher scores reflecting poorer quality of life. In the current study, the scale demonstrated excellent internal consistency, with a Cronbach's α coefficient of 0.922.

3.4. Data collection

Participants were recruited from 3 tertiary grade-A hospitals in China. The investigation was conducted with the prior approval of the hospital administrators. First, the chief nurse assisted in conducting the inquiry with the prior consent of the hospital

management, using the department as the unit. Second, before the investigation, the researchers informed the participants about the study's aim, importance, and privacy. The study participants were free to decline or leave at any time, and the information collected will only be used for academic research and not for profit. In this manner, the informed consent form was signed before the research, and the participant's consent was acquired. In-person, the surveys were completed in around 15–20 minutes, and participants were asked to fill in the questionnaires face-to-face, after which they were immediately collected and verified. In this study, a total of 451 questionnaires were collected within the specified time, and after excluding 19 invalid questionnaires, a total of 432 valid questionnaires were obtained, with an effective recovery rate of 95.8%.

3.5. Statistical analysis

Two researchers independently recorded and analyzed the raw data using Epidata 3.1, IBM SPSS 26.0, and AMOS 24.0. Normality of continuous variables (SRF, PM, HPB, QoL scores) was assessed using skewness, kurtosis, Kolmogorov-Smirnov test, and Q-Q plots. The skewness values ranged from -0.67 to 0.58 , and kurtosis from -0.42 to 0.53 , all within acceptable thresholds ($|\text{skewness}| < 2$, $|\text{kurtosis}| < 7$). Homogeneity of variance for group comparisons was checked using Levene's test. For one-way ANOVA, when variances were unequal, Welch's ANOVA was used, followed by Games-Howell post-hoc tests. For multiple comparisons in one-way ANOVA involving more than two groups (e.g., comparisons of QoL across age groups, educational levels, or comorbidity categories), the Bonferroni correction was applied to adjust the significance level to α/k , where k is the number of pairwise comparisons. Adjusted p -values are reported as significant when $p_{\text{corrected}} < 0.05$. Pearson's correlation was used after confirming linearity and absence of outliers. Multiple linear regression used the enter method, with residuals checked for normality (P-P plot) and independence (Durbin-Watson = 1.92). Variance inflation factor (VIF) and tolerance were used to assess multicollinearity (all VIF < 5). Structural equation modeling (SEM) was performed using AMOS 24.0 with maximum likelihood estimation. Model identification was confirmed by positive degrees of freedom ($df = 96$) and convergence after 7 iterations. The bias-corrected bootstrap (5000 resamples) was used to test indirect effects. To assess the significance of mediation effects, the bias-corrected (BC) bootstrap method with 5,000 resamples was applied to assess the significance of mediation effects, and statistical significance was set at $p < 0.05$ (two-tailed).

3.6. Ethics considerations

The study protocol was approved by the Medical Ethics Committee of Hengyang Central Hospital (Approval No. HYZXY-2025-06-018, approved in June 2025). Written informed consent was obtained from each participant. All procedures were performed per the principles of the Declaration of Helsinki and relevant guidelines and regulations in China. After obtaining permission from hospital administrators, the researchers approached the participants with the help of the head nurses. The participants were given the right to refuse or withdraw from this study. The questionnaires were designed to ensure anonymity and confidentiality, with no identifying marks, names, or numbers linked to the participants. The data obtained were only used for academic research and not commercial purposes.

4. Results

4.1. Demographic characteristics of T2DM patients

The study cohort consisted of 64.4% male ($n = 278$) and 35.6% female ($n = 154$) participants, with additional demographic characteristics detailed in [Table 1](#).

4.2. Scores of quality of life, self-regulatory fatigue, personal mastery, and health-promoting behaviors among T2DM patients

The total score of quality of life was 69.41 ± 20.89 , with a mean item score of 2.57 ± 0.77 (where a lower mean score indicates better quality of life, as the DS-QoLS is negatively scored). Among the four dimensions of QoL, "social relations" had

Table 1. Demographic characteristics and univariate analysis of quality of life among T2DM patients (n=432, M±SD).

Items	Number [n (%)]	Quality of Life total score	Physical function	Psychological/ Spiritual	Social relations	Treatment
Gender						
Male	278(64.4)	64.11±17.32	27.29±8.61	20.07±7.97	9.70±3.87	7.05±2.87
Female	154(35.6)	72.34±22.12	31.18±11.91	21.80±8.41	11.26±4.17	8.10±3.41
<i>t</i>		4.272	3.91	2.082	3.807	3.409
<i>P</i>		0.000**	0.000**	0.038*	0.000**	0.001**
Age (years)						
18~<45	72(16.7)	61.03±14.23	25.38±4.50	18.71±7.93	10.10±3.97	6.85±2.82
45~<60	84(19.4)	66.30±18.60	28.54±9.25	19.99±7.74	10.27±4.08	7.50±3.08
≥60	276(63.9)	72.54±22.28	31.33±12.28	22.19±8.39	10.99±4.17	8.03±3.38
<i>F</i>		10.239	9.386	6.273	1.901	4.073
<i>P</i>		0.000**	0.000**	0.002**	0.151	0.018*
BMI (kg/m²)						
<18.5	72(16.7)	69.17±20.63	29.25±10.46	21.49±8.30	10.60±4.17	7.83±3.39
18.5~23.9	108(25.0)	64.17±16.64	26.93±7.97	19.90±8.19	9.94±3.94	7.40±2.81
24.0~27.9	216(50.0)	69.75±20.99	30.12±11.13	21.19±8.09	10.77±4.03	7.67±3.25
≥28	36(8.3)	83.53±25.82	37.50±15.03	24.36±9.17	12.78±4.59	8.89±4.07
<i>F</i>		8.156	8.923	2.691	4.383	1.953
<i>P</i>		0.000**	0.000**	0.046*	0.005**	0.120
Educational level						
Primary school and below	48(11.1)	80.17±23.35	35.31±13.71	24.15±7.68	11.83±4.44	8.88±3.52
Junior high school	108(25.0)	70.23±22.21	30.27±11.93	20.82±8.41	11.06±4.15	8.08±3.52
Senior high school or technical secondary school	168(38.9)	69.82±20.94	29.45±10.85	21.99±8.28	10.70±4.27	7.68±3.15
College and above	108(25.0)	63.16±15.78	27.39±7.68	18.97±7.96	9.85±3.58	6.95±2.86
<i>F</i>		7.903	6.075	5.355	3.036	4.604
<i>P</i>		0.000**	0.000**	0.001**	0.029*	0.003**
Marital status						
Unmarried	52(12.0)	64.15±17.25	27.23±8.31	19.23±8.01	10.40±3.82	7.29±3.27
Married	315(72.9)	68.12±19.38	29.11±10.26	20.98±8.10	10.47±4.03	7.56±3.08
Divorced	27(6.3)	74.59±25.51	31.89±12.65	22.07±8.40	12.22±4.27	8.41±3.62
Widowed	38(8.8)	83.53±27.46	37.42±15.27	24.92±9.23	11.95±4.90	9.24±3.99
<i>F</i>		8.225	8.142	3.775	2.832	3.761
<i>P</i>		0.000**	0.000**	0.011*	0.038*	0.011*
Living conditions						
Living alone	47(10.9)	80.47±25.03	35.4±14.21	24.15±8.22	12.15±3.79	8.77±3.89
Living with children	107(24.8)	67.37±19.00	28.59±9.56	20.48±8.36	10.86±4.20	7.45±3.01
Living with spouse only	205(47.5)	65.94±17.99	28.05±9.14	20.56±8.13	9.85±3.88	7.48±3.00
Living with parents	49(11.3)	73.18±25.63	31.51±12.86	21.71±8.42	11.82±4.45	8.14±3.59
Others	24(5.6)	78.71±22.87	35.54±14.76	22.75±8.33	12.17±4.33	8.25±4.06
<i>F</i>		6.909	6.968	2.277	5.546	2.059
<i>P</i>		0.000	0.000	0.060	0.000	0.085
Residence						
Rural	84(19.4)	79.39±25.42	35.19±14.22	23.48±8.37	11.63±4.52	9.10±3.78
Urban	348(80.6)	66.99±18.91	28.49±9.65	20.63±8.19	10.48±4.00	7.40±3.04
<i>F</i>		4.198	4.097	2.848	2.311	3.824
<i>P</i>		0.000**	0.000**	0.005**	0.021*	0.000**

(Continued)

Table 1. (Continued)

Items	Number [n (%)]	Quality of Life total score	Physical function	Psychological/ Spiritual	Social relations	Treatment
Employment status						
Employed	79(18.3)	61.70 ± 13.34	25.52 ± 4.39	19.03 ± 7.69	10.23 ± 3.80	6.92 ± 2.90
Retired	310(71.8)	69.86 ± 21.43	30.18 ± 11.33	21.27 ± 8.30	10.62 ± 4.15	7.78 ± 3.25
Unemployed	43(9.9)	80.33 ± 23.14	34.81 ± 14.19	24.49 ± 8.32	12.16 ± 4.33	8.86 ± 3.65
F		11.897	11.136	6.26	3.312	5.13
P		0.000**	0.000**	0.002**	0.037*	0.006**
Per capita monthly income (RMB)						
<3000	156(36.1)	73.63 ± 22.87	31.87 ± 13.04	22.46 ± 8.20	11.01 ± 4.37	8.29 ± 3.41
3000~<50000	204(47.2)	68.95 ± 20.62	29.50 ± 10.54	20.93 ± 8.39	10.85 ± 4.03	7.67 ± 3.20
≥5000	72(16.7)	61.54 ± 13.80	26.11 ± 4.75	19.14 ± 7.83	9.61 ± 3.71	6.68 ± 2.83
F		8.632	7.063	4.202	3.115	6.21
P		0.000**	0.001**	0.016*	0.045*	0.002**
Medical insurance payment method						
Self-payment	25(5.8)	66.58 ± 18.68	28.45 ± 9.67	20.41 ± 8.12	10.26 ± 3.92	7.47 ± 3.06
New rural cooperative medical care	83(19.2)	71.60 ± 21.54	30.31 ± 11.43	22.29 ± 7.96	11.25 ± 4.43	7.75 ± 3.29
Urban medical insurance	275(63.7)	81.20 ± 27.47	34.84 ± 14.42	25.12 ± 9.65	11.64 ± 4.84	9.60 ± 3.69
Others	49(11.3)	75.51 ± 24.37	33.88 ± 13.57	21.63 ± 8.53	11.78 ± 4.06	8.22 ± 3.79
F		6.25	5.614	3.265	3.134	3.791
P		0.000**	0.001**	0.021*	0.025*	0.011*
Drinking status						
Never	155(35.9)	65.17 ± 18.32	28.05 ± 9.19	19.93 ± 8.20	9.99 ± 3.81	7.20 ± 2.96
Occasionally	168(38.9)	70.03 ± 20.90	30.05 ± 10.88	21.44 ± 8.29	10.86 ± 4.16	7.68 ± 3.26
Frequently	109(25.2)	74.47 ± 23.15	31.87 ± 13.07	22.57 ± 8.24	11.47 ± 4.37	8.56 ± 3.52
F		6.63	3.987	3.414	4.398	5.712
P		0.001**	0.019*	0.034*	0.013*	0.004**
Smoking status						
Never	60(13.9)	60.92 ± 14.56	25.83 ± 5.35	18.82 ± 7.78	9.40 ± 3.89	6.87 ± 2.73
Occasionally	70(16.2)	69.04 ± 20.35	30.30 ± 11.80	20.84 ± 8.01	10.60 ± 4.15	7.30 ± 3.30
Frequently	302(69.9)	71.18 ± 21.70	30.46 ± 11.49	21.73 ± 8.39	10.98 ± 4.13	8.00 ± 3.32
F		6.194	4.592	3.196	3.756	3.803
P		0.002**	0.011*	0.042*	0.024*	0.023*
Disease duration (years)						
<1	24(5.6)	58.42 ± 11.65	25.42 ± 6.45	17.42 ± 7.51	9.58 ± 3.57	6.00 ± 2.70
1~<6	132(30.6)	68.23 ± 20.19	29.53 ± 10.57	20.56 ± 7.98	10.45 ± 4.02	7.70 ± 3.23
6~<10	192(44.4)	68.35 ± 20.14	29.18 ± 10.37	21.06 ± 8.38	10.46 ± 4.04	7.65 ± 3.11
≥10	84(19.4)	76.80 ± 23.61	32.85 ± 13.31	23.51 ± 8.30	11.98 ± 4.43	8.46 ± 3.61
F		6.240	3.712	4.213	3.716	3.792
P		0.000**	0.012*	0.006**	0.012*	0.010*
Number of comorbidities						
None	28(6.5)	54.79 ± 15.30	24.43 ± 6.90	15.68 ± 6.56	8.50 ± 4.02	6.18 ± 2.06
1~2	276(63.9)	69.07 ± 20.69	29.56 ± 10.71	21.08 ± 8.32	10.68 ± 4.12	7.75 ± 3.24
≥3	128(29.6)	73.33 ± 21.02	31.46 ± 11.96	22.60 ± 8.09	11.23 ± 4.03	8.03 ± 3.43
F		9.508	4.950	8.338	5.152	3.769
P		0.000**	0.007**	0.000**	0.006**	0.024*

(Continued)

Table 1. (Continued)

Items	Number [<i>n</i> (%)]	Quality of Life total score	Physical function	Psychological/ Spiritual	Social relations	Treatment
Treatment modality						
Oral medication	72(16.7)	70.36±21.20	30.06±11.66	22.31±7.91	10.61±4.22	7.39±3.29
Insulin	168(38.9)	70.14±21.88	30.08±11.61	21.27±8.37	10.87±4.15	7.92±3.34
Oral+ Insulin	132(30.6)	68.65±20.28	29.61±10.41	20.94±8.50	10.27±4.01	7.83±3.20
Others	60(13.9)	67.87±19.32	29.05±9.87	20.13±8.09	11.28±4.20	7.40±3.16
<i>F</i>		0.283	0.155	0.803	0.976	0.690
<i>P</i>		0.837	0.927	0.493	0.404	0.558
Glycemic control status						
Poor	48(11.1)	84.85±25.99	37.13±14.60	25.44±8.01	12.63±4.37	9.67±3.94
Moderate	108(25)	68.92±20.86	29.46±10.63	20.76±8.55	10.84±4.28	7.85±3.27
Good	276(63.9)	66.91±18.75	28.64±9.93	20.61±8.05	10.31±3.94	7.34±3.01
<i>F</i>		16.186	12.885	7.331	6.678	10.955
<i>P</i>		0.000**	0.000**	0.001**	0.001**	0.000**
Whether combined with other chronic diseases						
Yes	394(91.2)	70.36±21.21	30.20±11.31	21.47±8.25	10.82±4.12	7.87±3.27
No	38(8.8)	59.50±14.07	25.58±5.57	18.21±8.29	9.42±4.03	6.29±2.78
<i>t</i>		4.311	4.325	2.326	2.009	2.876
<i>P</i>		0.000**	0.000**	0.020*	0.045*	0.004**
Self-care ability						
Fully independent	132(30.6)	64.79±17.42	27.30±8.62	20.17±7.79	10.05±3.94	7.27±2.93
Partially independent	252(58.3)	68.90±20.55	29.63±10.73	20.99±8.45	10.56±4.07	7.72±3.17
Completely dependent	48(11.1)	84.73±24.51	37.48±14.49	25.00±7.90	13.23±4.05	9.02±4.20
<i>F</i>		17.442	16.212	6.299	11.294	5.151
<i>P</i>		0.000**	0.000**	0.002**	0.000**	0.006**

Note: ** $p < 0.01$. For post-hoc pairwise comparisons following one-way ANOVA, the Bonferroni correction was applied. Adjusted p-values are reported; significance was set at $p_{\text{corrected}} < 0.05$. For the Diabetes-Specific Quality of Life Scale (DS-QoLS) used in this table, a lower score indicates better quality of life.

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the highest average score (2.68 ± 1.03), while “physical function” had the lowest average score (2.48 ± 0.92). The average scores for the other dimensions were as follows: “Psychological/Spiritual” (2.65 ± 1.04), and “Treatment” (2.58 ± 1.09).

The total score of self-regulatory fatigue was 38.63 ± 12.84 , with an overall mean of 2.41 ± 0.80 . Among the three dimensions of the SRFS, “emotional control” had the highest average score (2.67 ± 1.05), while “behavioral control” had the lowest (2.00 ± 0.96). The average scores for the “cognitive control” dimensions was 2.65 ± 1.20 .

The total score of personal mastery was 26.12 ± 7.62 , and the average score was 3.73 ± 1.09 .

The total score of health-promoting behaviors was 97.69 ± 20.87 , and the average score was 3.48 ± 0.95 . Among the six dimensions of the DHPS, “stress management” had the highest average score (3.57 ± 0.98), while “exercise” had the lowest (3.44 ± 1.09).

The scores of QoL, SRF, PM, and HPB were shown in [Table 2](#).

All assumptions for parametric tests (normality, homogeneity of variance, linearity, absence of multicollinearity, independence of residuals) were checked and met, as detailed in the Statistical analysis section. The skewness and kurtosis values for the total scores of key variables were as follows: SRF total score (skewness = 0.28, kurtosis = -0.35), PM total score (skewness = -0.67, kurtosis = 0.42), HPB total score (skewness = -0.41, kurtosis = 0.53), and QoL total score

Table 2. The scores of quality of life, self-regulatory fatigue, personal mastery, and health-promoting behaviors among T2DM patients ($n=432$, $M \pm SD$).

Items	Number of items	Score range	Dimension score	Average Item Score	Ranking
Quality of Life total score	27	27 ~ 135	69.41 $\pm$ 20.89	2.57 $\pm$ 0.77	
Physical function	12	12 ~ 60	29.79 $\pm$ 11.00	2.48 $\pm$ 0.92	4
Psychological/Spiritual	8	8 ~ 40	21.18 $\pm$ 8.29	2.65 $\pm$ 1.04	2
Social relations	4	4 ~ 20	10.70 $\pm$ 4.13	2.68 $\pm$ 1.03	1
Treatment	3	3 ~ 15	7.73 $\pm$ 3.26	2.58 $\pm$ 1.09	3
Self-Regulatory Fatigue total score	16	18 ~ 74	38.63 $\pm$ 12.84	2.41 $\pm$ 0.80	
Cognitive control	6	6 ~ 25	13.26 $\pm$ 6.01	2.65 $\pm$ 1.20	2
Behavioral control	5	6 ~ 30	12.00 $\pm$ 5.78	2.00 $\pm$ 0.96	3
Emotional control	5	5 ~ 25	13.37 $\pm$ 5.26	2.67 $\pm$ 1.05	1
Health-Promoting Behaviors total score	28	28 ~ 140	97.69 $\pm$ 20.87	3.48 $\pm$ 0.95	
Exercise	7	7 ~ 35	24.38 $\pm$ 6.61	3.44 $\pm$ 1.09	6
Risk avoidance	7	7 ~ 35	24.09 $\pm$ 7.61	3.51 $\pm$ 1.09	3
Stress management	5	5 ~ 25	17.56 $\pm$ 5.47	3.57 $\pm$ 0.98	1
Life appreciation	3	3 ~ 15	10.72 $\pm$ 2.93	3.46 $\pm$ 0.89	5
Health responsibility	3	3 ~ 15	10.39 $\pm$ 2.68	3.52 $\pm$ 1.02	2
Healthy diet	3	3 ~ 15	10.55 $\pm$ 3.06	3.48 $\pm$ 0.95	4
Personal Mastery total score	7	8 ~ 34	26.12 $\pm$ 7.62	3.73 $\pm$ 1.09	

Note: For the Diabetes-Specific Quality of Life Scale (DS-QoLS), a lower score indicates better quality of life.

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(skewness = 0.58, kurtosis = -0.22). These values confirm that the data met the assumption of approximate univariate normality.

4.3. Univariate analysis of quality of life among T2DM patients

The result showed that gender, age, BMI, educational level, marital status, living conditions, residence, employment status, per capita monthly income, medical insurance payment method, drinking status, smoking status, disease duration, number of comorbidities, glycemic control status, whether combined with other chronic diseases, and self-care ability as statistically significant factors associated with quality of life among T2DM patients (all $p < 0.05$), as shown in [Table 1](#).

4.4. Correlations among quality of life, self-regulatory fatigue, personal mastery and health-promoting behaviors among T2DM patients

As detailed in [Table 3](#), the total QoL score (where a higher score indicates poorer QoL) was positively correlated with the total SRF score ($r = 0.581$, $p < 0.01$, 95% CI [0.52, 0.64]). This positive correlation aligns with the scale's direction, indicating that higher self-regulatory fatigue is associated with a poorer quality of life. In contrast, the QoL total score was negatively correlated with PM ($r = -0.557$, $p < 0.01$, 95% CI [-0.62, -0.49]) and HPB ($r = -0.613$, $p < 0.01$, 95% CI [-0.67, -0.55]). Because a lower QoL score denotes better quality of life, these negative correlations signify that higher levels of personal mastery and more frequent engagement in health-promoting behaviors are associated with significantly better QoL. Furthermore, the analysis revealed a strong, positive correlation between PM and HPB ($r = 0.607$, $p < 0.01$), which provides the empirical foundation for testing the hypothesized chain-mediating effect. All subscale correlations were significant and in the expected directions.

Table 3. Correlations among quality of life, self-regulatory fatigue, personal mastery, and health-promoting behaviors among T2DM patients (n=432, r).

Items	Quality of life total score	Physical function	Psychological/ Spiritual	Social relations	Treatment
Self-regulatory fatigue total score	0.581**	0.525**	0.463**	0.455**	0.537**
Cognitive control	0.472**	0.461**	0.422**	0.369**	0.447**
Behavioral control	0.358**	0.336**	0.298**	0.312**	0.328**
Emotional control	0.439**	0.415**	0.386**	0.401**	0.421**
Health-promoting behaviors total score	-0.613**	-0.536**	-0.492**	-0.501**	-0.562**
Exercise	-0.523**	-0.473**	-0.451**	-0.395**	-0.408**
Risk avoidance	-0.508**	-0.482**	-0.463**	-0.376**	-0.411**
Stress management	-0.496**	-0.447**	-0.440**	-0.347**	-0.373**
Life appreciation	-0.479**	-0.439**	-0.438**	-0.361**	-0.382**
Health responsibility	-0.431**	-0.415**	-0.427**	-0.315**	-0.348**
Healthy diet	-0.595**	-0.498**	-0.441**	-0.409**	-0.487**
Personal mastery total score	-0.557**	-0.519**	-0.523**	-0.498**	-0.536**

Note: ** $p < 0.01$. For the Diabetes-Specific Quality of Life Scale (DS-QoLS) used in this table, a lower score indicates better quality of life.

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4.5. Multiple linear regression analysis of quality of life among T2DM patients

In this study, the quality of life of T2DM patients served as the dependent variable. Independent variables comprised seventeen statistically significant sociodemographic and clinical characteristics identified through univariate analysis, along with variables demonstrating statistical significance in correlation analysis. Variable coding specifications are detailed in [Table 4](#), with continuous variables retained in their original continuous form. A multiple linear regression model was constructed to identify significant correlates of quality of life among T2DM patients, ultimately incorporating 10 variables into the final equation.

To assess potential multicollinearity among the independent variables, we examined the variance inflation factor (VIF) and tolerance. The results showed that VIF values for all predictors ranged from 1.12 to 3.85, and tolerance values ranged from 0.26 to 0.89. These values are well below the commonly accepted threshold ($VIF < 5$), indicating that multicollinearity did not significantly influence the stability of our regression model. The results of multiple linear regression analysis revealed ten key variables significantly associated with quality of life among T2DM patients ($p < 0.05$): age, educational level, medical insurance payment method, smoking status, glycemic control status, number of comorbidities, self-care ability, self-regulatory fatigue, personal mastery, and health-promoting behaviors, indicating that these were the main factors associated with quality of life among T2DM patients, explaining 41.8% of the total variation, as shown in [Table 5](#).

4.6. Mediating roles of personal mastery and health-promoting behaviors between self-regulatory fatigue and quality of life

The structural equation model, with self-regulatory fatigue as the independent variable, quality of life as the dependent variable, and personal mastery and health-promoting behaviors as mediators, showed a good fit to the data without any additional covariance paths or correlated residual errors beyond the hypothesized structural model (i.e., no post-hoc model modifications were made) [22,23]: $\chi^2/df = 1.906$ (cutoff < 3), RMSEA = 0.046 (cutoff < 0.08), GFI = 0.932, AGFI = 0.912, TLI = 0.963, CFI = 0.968, IFI = 0.969 (cutoff for all > 0.90). These indices collectively indicate that our hypothesized mediation model is highly consistent with the observed data. The corresponding path diagram is presented in [Fig 2](#). Despite

Table 4. Assignment of independent variables ($n=432$).

Independent Variables	Assignment
Age	Dummy variables were set with "18~<45" as the reference: "45~<60" = (0,1,0), "≥60" = (0,0,1)
Educational level	Dummy variables were set with "Primary school and below" as the reference: "Junior high school" = (0,1,0,0), "Senior high school or technical secondary school" = (0,0,1,0), "College and Above" = (0,0,0,1)
Medical insurance payment method	Dummy variables were set with "Self-payment" as the reference: "New rural cooperative medical care" = (0,1,0,0), "Urban medical insurance" = (0,0,1,0), "Others" = (0,0,0,1)
Self-care ability	Dummy variables were set with "Fully independent" as the reference: "Partially independent" = (0,1,0), "Completely dependent" = (0,0,1)
Glycemic control status	Dummy variables were set with "Poor" as the reference: "Moderate" = (0,1,0), "Good" = (0,0,1)
Smoking status	Dummy variables were set with "Never" as the reference: "Occasionally" = (0,1,0), "Frequently" = (0,0,1)
Number of comorbidities	Dummy variables were set with "None" as the reference: "1~2" = (0,1,0), "≥3" = (0,0,1)
Self-regulatory fatigue total score	Original entry
Personal mastery total score	Original entry
Health-promoting behaviors total score	Original entry

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Table 5. Multiple linear regression analysis of quality of life among T2DM patients ($n=432$).

Dependent variable	Independent variables	Unstandardized Coefficient (B)	SE	Standardized Coefficient (Beta)	95% CI for B	t	p	VIF
Quality of life	Constant	3.996	0.265	—	—	15.060	0.000	—
	Age	0.110	0.037	0.108	[0.035, 0.181]	3.001	0.003	2.23
	Educational level	-0.071	0.024	-0.087	[-0.312, -0.193]	-2.940	0.003	1.56
	Medical insurance payment method	-0.097	0.035	-0.088	[-0.351, -0.205]	-2.743	0.006	2.11
	Smoking status	0.126	0.037	0.118	[0.028, 0.176]	3.435	0.001	1.69
	Glycemic control status	-0.137	0.036	-0.121	[-0.417, -0.298]	-3.789	0.000	1.81
	Number of comorbidities	0.084	0.041	0.060	[0.013, 0.208]	2.067	0.039	3.85
	Self-care ability	0.178	0.039	0.142	[0.026, 0.327]	4.554	0.000	1.73
	Self-regulatory fatigue	0.175	0.033	0.181	[0.021, 0.217]	5.262	0.000	2.28
	Personal mastery	-0.203	0.023	-0.285	[-0.035, -0.181]	-8.676	0.000	1.45
	Health-promoting behaviors	-0.394	0.040	-0.363	[-0.442, -0.284]	-9.902	0.000	1.12

Note: ** $p < 0.01$, $R = 0.656$, $R^2 = 0.431$, adjusted $R^2 = 0.418$, $F = 112.518$, $p < 0.001$.

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the significant correlation between PM and HPB, an assessment of multicollinearity showed that the variance inflation factor (VIF) for all predictors in the model was below the common threshold of 5 (ranging from 1.12 to 3.85, as reported in the multiple regression analysis), indicating that collinearity did not significantly distort the stability of the path coefficient estimations.

Bootstrap analysis indicated that the hypothesized total effect of SRF on QoL was 0.845, comprising a direct effect of 0.362 and a total indirect effect of 0.483. The total indirect effect accounted for 57.2% of the total effect.

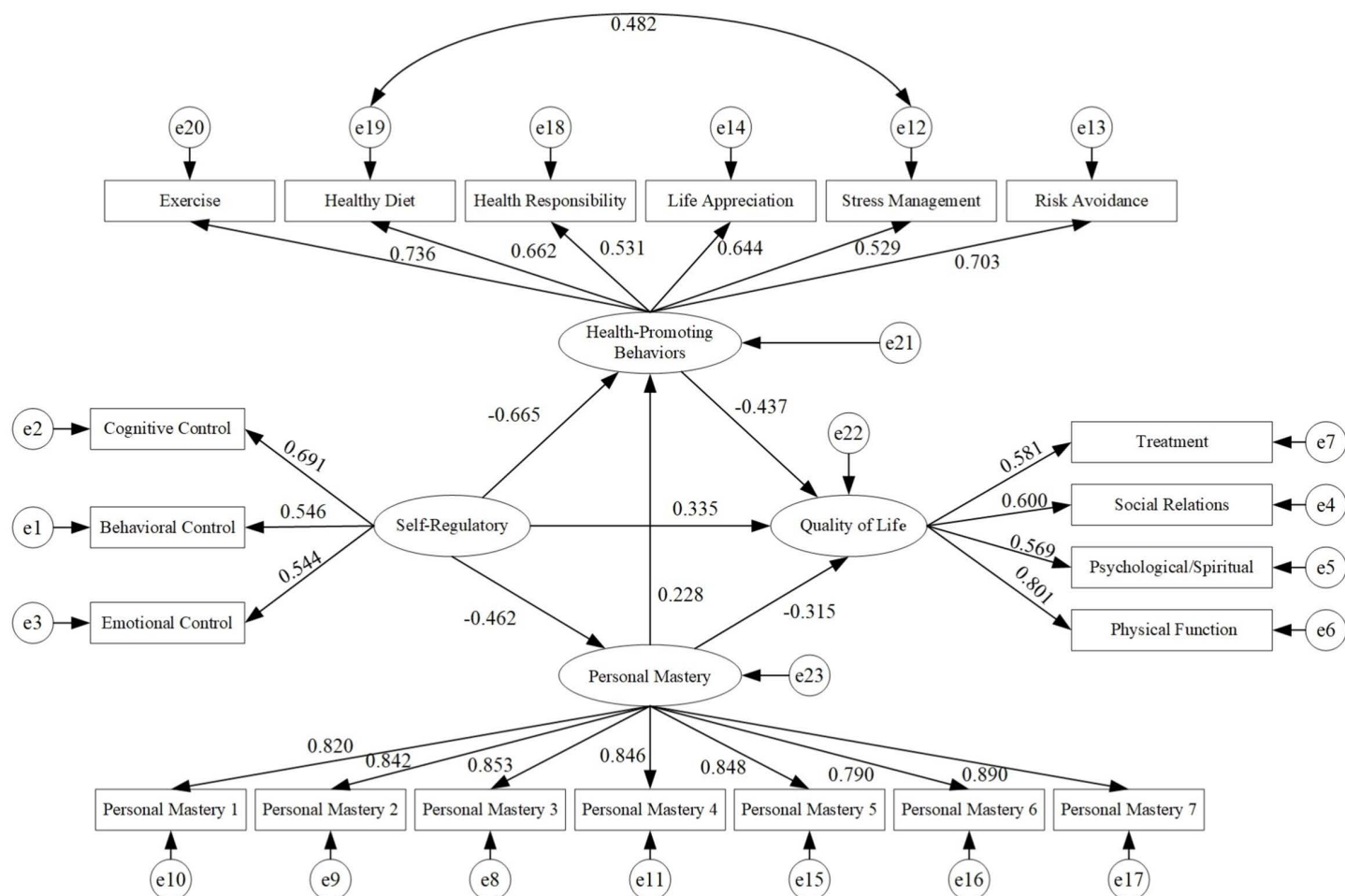


Fig 2. Multiple mediating effect model of personal mastery and health-promoting behaviors in the relationship between self-regulatory fatigue and quality of life.

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To provide a detailed decomposition of the mediation, we calculated each specific indirect pathway's contribution. Relative to the total effect (0.845), the specific indirect effects were as follows: via PM only (0.146, contributing 17.3%), via HPB only (0.291, contributing 34.4%), and via the chain of PM→HPB (0.046, contributing 5.4%). Relative to the total indirect effect (0.483), these same pathways accounted for 30.2% (PM only), 60.2% (HPB only), and 9.5% (PM→HPB chain), respectively. For consistency with our initial analytical framework and to aid comparison with prior studies that decompose the indirect effect, we report the proportions relative to the total indirect effect in the main text. Thus, the mediating effect of personal mastery accounted for 30.2% of the total indirect effect, while that of health-promoting behaviors explained 60.2%. The chain mediating effect contributed 9.5% of the total indirect effect.

Overall, the total mediating effect accounted for 57.2% of the total effect, as shown in the [Table 6](#).

5. Discussion

5.1. Status quo of quality of life among T2DM patients

The overall quality of life among T2DM patients in this study was moderate, with an average item score of approximately 2.57 out of 5 (with lower scores denoting better QoL). This finding aligns with previous reports by Wang et al. [24], Cheng

Table 6. Decomposition of direct, indirect, and total effects in the mediation model ($n=432$).

Effect type	Pathway	Effect size	95% CI (LCI, UCI)	Proportion of total effect (%)	Proportion of total indirect effect (%)
Direct effect	SRF → QoL	0.362	[0.221, 0.503]	42.8%	—
Total Indirect Effect	SRF → PM/HPB → QoL	0.483	[0.351, 0.615]	57.2%	100.0%
Specific indirect effects	SRF → PM → QoL	0.146	[0.089, 0.203]	17.3%	30.2%
	SRF → HPB → QoL	0.291	[0.182, 0.400]	34.4%	60.2%
	SRF → PM → HPB → QoL (Chain)	0.046	[0.018, 0.074]	5.4%	9.5%
Total effect	SRF → QoL	0.845	[0.713, 0.977]	100.0%	—

Note: All indirect effects were significant at $p < 0.01$ (two-tailed), as determined by bias-corrected (BC) bootstrap 95% confidence intervals. The number of bootstrap resamples was 5,000.

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et al. [25], and Cui et al. [26]. The moderate QoL level likely reflects the clinical reality that while T2DM imposes certain limitations on physical and psychological functioning, most patients do not experience profound functional loss. Instead, they often adapt to a state of living with the disease and partial functional impairment.

Notably, participants reported relatively better QoL in the social relations and psychological/spiritual domains. The relatively higher score in social relations may be attributed to both the nature of the items and the supportive responses elicited by chronic disease management. The items in this domain focus on general aspects of social interaction and perceived support, which are not inherently restricted by T2DM. Moreover, the long-term management context often mobilizes support from family and friends (e.g., dietary reminders, emotional encouragement), potentially enhancing patients' subjective social well-being and buffering the disease's negative impact. The psychological/spiritual domain also ranked second, possibly due to variable individual coping mechanisms. Although T2DM management can induce stress, anxiety, or depression, some patients effectively mitigate these through cognitive reappraisal and leveraging social support, introducing individual variability that attenuates a uniformly negative effect of the disease on this domain.

In contrast, participants reported relatively poorer QoL in the treatment and physical function domains. The intermediate score in the treatment domain likely represents a balance between the burden of ongoing self-management and necessary treatment adherence. While the demands of continuous medication and glucose monitoring contribute to psychological and practical burden, awareness of long-term complications may sustain basic compliance, preventing a more substantial decline in this domain's score. The physical function domain received the lowest score, which directly reflects the core pathophysiology of T2DM. This domain comprehensively assesses functional indicators like physical activity, sleep quality, and pain—all susceptible to diabetes complications such as neuropathy, fatigue, and vascular issues. Chronic hyperglycemia and insulin resistance can also lead to end-organ damage (e.g., cardiac or renal impairment), further limiting physical capacity and significantly lowering this QoL dimension.

These findings have several implications for clinical nursing practice. First, interventions should prioritize improving physical status through strategies such as optimizing glycemic control, actively managing pain (e.g., from neuropathy), and implementing structured rehabilitation programs. Second, efforts should reinforce the protective role of social relations by encouraging family and community involvement in disease management. Finally, to address the treatment-related burden, nursing care could incorporate hierarchical psychological support and explore ways to simplify complex treatment regimens where possible, thereby helping to balance the demands of adherence with patient well-being.

5.2. Factors associated with quality of life among T2DM patients

5.2.1. Age. The results of this study demonstrate that the QoL in T2DM patients significantly declines with age, with distinct hierarchical characteristics observed in the core influencing factors and the extent of QoL impairment across

different age groups. These findings are consistent with those reported by Zhang et al. [27]. The underlying reasons can be summarized as follows:

- (1) In the 18–<45 age group, patients' islet function remains relatively compensatory, insulin resistance is manageable, and complications are predominantly early-stage microangiopathy that does not immediately impact physical function. The primary source of stress in this group arises from role conflicts between career development and family responsibilities, leaving fragmented time for disease management. Nevertheless, their relatively high disease awareness and good self-management compliance enable them to maintain a favorable QoL under standardized interventions. Their psychological burden mainly stems from concerns about the disease's impact on marriage, childbearing, and career advancement.
- (2) In the 45–<60 age group, patients experience a marked age-related decline in islet secretory function, increased insulin resistance, greater glycemic variability, and a higher susceptibility to multiple complications, including diabetic nephropathy, retinopathy, and peripheral neuropathy. This group frequently presents with concomitant chronic conditions such as hypertension and hyperlipidemia, and the risk of drug interactions and adverse reactions due to polypharmacy is elevated, further compromising treatment adherence. In addition, social support gradually diminishes, and the interaction between physical symptoms and negative psychological emotions contributes to a marked increase in the incidence of depression, leading to a reduced QoL.
- (3) In the ≥ 60 age group, islet function is often in a state of severe decline, glycemic regulation is markedly impaired, and disabling complications such as diabetic foot and advanced diabetic nephropathy are common. Some patients also experience cognitive decline, resulting in a complete loss of self-management ability and a high degree of dependence on caregivers. Furthermore, the overall decline in systemic physiological functions significantly reduces tolerance to hypoglycemic agents and treatments for complications, prompting clinical decisions to shift toward palliative interventions. Physical function is severely restricted in this group; some patients require prolonged bed rest and experience a total loss of autonomy in daily life, culminating in the lowest QoL.

5.2.2. Educational level. The findings of this study indicate a positive correlation between educational level and QoL in T2DM patients; specifically, higher educational attainment is associated with better QoL. This result aligns with the findings of Yu [28]. The underlying reasons are as follows:

- (1) Patients with higher educational levels possess a more comprehensive and scientific understanding of the etiology, pathology, and treatment regimens of diabetes. They are better able to grasp core management concepts—such as blood glucose control targets and the interplay between diet, exercise, and medication—resulting in more standardized and sustained self-management behaviors (e.g., regular blood glucose monitoring, medication adherence, and adherence to dietary and exercise recommendations). Consequently, this group achieves significantly higher rates of blood glucose compliance. Research indicates that the blood glucose compliance rate exceeds 60% among patients with higher educational levels, compared with only approximately 30% among those with lower educational levels [29]. Moreover, each incremental increase in educational level is associated with an average 8%–10% increase in the rate of blood glucose control compliance [30]. As a result, these patients experience a lower incidence and severity of complications, which directly contributes to improved QoL.
- (2) Patients with higher educational levels are more proactive in seeking professional medical information—such as clinical guidelines and authoritative health resources—and demonstrate greater efficiency in utilizing healthcare services. They communicate more effectively with healthcare providers and are more willing to engage in disease management programs, including diabetes education courses and remote follow-up services, thereby enabling more precise use of medical resources. These patients also exhibit a stronger grasp of strategies for managing drug-related side effects

and implementing non-pharmacological interventions, leading to higher treatment adherence and efficacy, and reducing the risk of disease instability caused by improper self-management.

- (3) Patients with higher educational levels typically possess stronger skills in evaluating health information, which helps mitigate irrational beliefs, such as disease-related anxiety, and enhances their psychological resilience. In addition, they often benefit from broader social networks and more robust social support, alleviating the psychological burden associated with disease management. Studies have shown that the incidence of depression or anxiety is 20%–30% lower among patients with a high school education or above than among those with a primary school education or below [31]. This improved psychological state further contributes positively to both physical function and overall QoL.

5.2.3. Medical insurance payment method. This study found that patients with urban medical insurance reported the highest QoL scores, whereas self-paying patients reported the lowest, a finding consistent with the study by Li [32]. The underlying reasons can be summarized into three aspects.

- (1) Medical insurance alleviates the financial burden of treatment, whereas self-paying patients face substantial economic pressure. Diabetes requires long-term medication and regular follow-up; thus, financial constraints can directly affect treatment continuity. Urban medical insurance offers a high reimbursement rate and a comprehensive formulary that fully covers insulin, blood glucose monitoring, and complication screening. As a result, out-of-pocket expenses are low, reducing the likelihood of treatment interruption due to cost. Although the New Rural Cooperative Medical Scheme (NRCMS) covers some expenses, its reimbursement rate, cap, and formulary are generally less generous than those of urban medical insurance. For instance, in some regions, the outpatient reimbursement rate for chronic diseases under NRCMS ranges from 50% to 70%, which is lower than that of urban medical insurance, and reimbursement for care received outside primary facilities is limited. In contrast, self-paying patients bear all costs; the high price of medications and tests often forces them to reduce medication dosages or skip follow-up visits, leading to disease progression and decreased QoL.
- (2) Access to high-quality medical resources is better for patients with urban medical insurance than for rural and self-paying patients. The designated facilities for urban medical insurance are often high-tier institutions, such as Grade A tertiary hospitals, enabling patients to receive standardized care, including timely screening and intervention for diabetic complications (e.g., retinopathy and foot ulcers). Although patients covered by NRCMS can access care through primary healthcare networks, high-quality resources are concentrated in urban areas, making cross-level care both time-consuming and costly. Moreover, township health centers often have limited capacity for managing complications. Self-paying patients, constrained by financial pressures, tend to seek care at primary clinics or self-medicate, making it difficult to access professional multidisciplinary teams and advanced diagnostic tools—a situation that increases the risk of misdiagnosis and non-standardized treatment, further impairing QoL.
- (3) Urban medical insurance is associated with more standardized health management, whereas self-paying patients lack such support. Urban medical insurance is often linked to community health services, providing patients with structured management—including blood glucose monitoring guidance, dietary advice, and exercise interventions—to support glycemic control and reduce complications. Although NRCMS encourages more frequent medical visits through policies such as chronic disease cards, patients' awareness of reimbursement policies and disease management knowledge remains low, limiting treatment adherence. In the absence of the structured guidance provided by medical insurance, self-paying patients often delay seeking care due to cost concerns, presenting only when severe symptoms develop, at which point complications are already established—resulting in a marked decline in QoL.

5.2.4. Smoking status. This study revealed that, among T2DM patients, QoL was highest in those who had never smoked, followed by occasional smokers, and lowest in frequent smokers—a pattern consistent with the findings of Wang et al. [33]. The underlying reasons are as follows:

- (1) Smoking complicates glycemic control. Patients who never smoke are not exposed to exogenous substances that interfere with metabolism, allowing for more stable blood glucose levels. Smoking activates the sympathetic nervous system, induces insulin resistance, and reduces the efficacy of glucose-lowering medications. On average, frequent smokers have fasting blood glucose levels 1.8–2.5 mmol/L higher than non-smokers [34]. Occasional smoking causes only transient glycemic disruption, whereas frequent smoking leads to sustained interference and persistently elevated glycated hemoglobin, ultimately impairing QoL.
- (2) The risk of complications increases with smoking frequency [35]. Non-smokers largely avoid smoking-related vascular and organ damage, resulting in a very low incidence of complications. Frequent smoking damages the vascular endothelium, increasing the risk of coronary heart disease by 2.3-fold in diabetic patients compared with non-smokers, and accelerates the progression of nephropathy, with a significantly faster annual decline in renal function [36]. Although occasional smoking also has adverse effects on blood vessels and the liver, the cumulative damage is limited due to lower exposure frequency, and the associated complication risk is substantially lower than that in frequent smokers.
- (3) Physical and mental well-being are differentially affected by smoking status [37]. Non-smokers experience neither smoking-related physical discomfort nor the associated psychological burden, contributing to good QoL. Frequent smokers not only endure symptoms such as cough and liver discomfort but also face anxiety and depression driven by a high risk of complications and increased treatment costs. Depression, in turn, reduces treatment adherence, creating a vicious cycle. Although occasional smokers experience less psychological distress, mild anxiety may still arise from health concerns, and short-term physical discomfort can diminish their QoL.

5.2.5. Glycemic control status. The findings of this study indicate that better glycemic control is associated with better QoL, which is consistent with the results reported by Peng et al. [38]. This relationship can be attributed to the following factors. (1) Long-term effective glycemic management prevents sustained hyperglycemic damage to blood vessels and nerves, thereby substantially reducing the risk of complications. The UK Prospective Diabetes Study (UKPDS) demonstrated that early intensive glucose control reduces complications such as renal failure and blindness by 26%, and intensive glucose control with metformin lowers the risk of myocardial infarction by 31% [39]. In contrast, poor glycemic control increases the risk of complications such as retinopathy and diabetic foot, which directly impair daily functioning through vision loss and ambulatory difficulties. Effective glucose management helps avoid these issues and preserves normal physical function. (2) Patients with well-controlled blood glucose seldom require frequent adjustments to their treatment regimens; most can maintain stability through basic dietary management and minimal oral medication, without the burden of complex treatments such as insulin injections. (3) Poor glycemic control often leads to anxiety and depression due to concerns about disease progression, and the continuous, intensive effort required for glucose regulation may result in self-regulatory fatigue. A study involving 187 T2DM patients found that glycosylated hemoglobin levels were positively correlated with self-regulatory fatigue scores, and self-regulatory fatigue mediated 47.59% of the effect on QoL [40]. Stable blood glucose levels can enhance patients' confidence in treatment, reduce psychological burden and self-regulatory fatigue, create a virtuous cycle, and ultimately improve QoL.

5.2.6. Number of comorbidities. The results of this study indicate that a greater number of comorbidities is associated with poorer QoL among patients, which is consistent with the findings of Han et al. [41]. This association can be attributed to the following factors:

- (1) Diabetes itself contributes to vascular and neurological damage, and the presence of additional chronic conditions may exert a synergistic pathological effect (i.e., $1 + 1 > 2$). For instance, concurrent hypertension can exacerbate arteriosclerosis; the coexistence of diabetic foot increases the risk of limited mobility and impaired daily activities by more than fourfold; and comorbid stroke significantly heightens the risk of compromised self-care ability. As the number of comorbidities increases, patients' organ compensatory capacity progressively declines, accompanied by deteriorating physical function and comfort.

- (2) A higher burden of comorbidities is often associated with more complex treatment regimens and an elevated risk of drug–drug interactions. For example, patients with diabetes and concomitant nephropathy must carefully balance glycemic control with renal protection while avoiding nephrotoxic medications. Similarly, in patients with diabetes and comorbid depression, interactions between antidepressants and hypoglycemic agents may compromise therapeutic efficacy. In addition, economic burden tends to escalate with the number of comorbidities. A survey involving 7,082 patients in China [42] reported that 81.45% had comorbidities, with their associated diagnosis and treatment costs substantially exceeding those of patients without comorbidities. Each additional comorbidity incurs additional expenses for relevant diagnostic tests and medications. Such financial strain may lead patients to reduce healthcare spending, thereby exacerbating disease progression and further diminishing QoL.
- (3) The persistent distress associated with multiple chronic conditions is likely to precipitate negative emotional states. Research indicates that comorbid conditions are associated with a 16.28% reduction in patients' emotional functioning [43]. Moreover, a higher comorbidity burden is linked to heightened anxiety and stress related to concerns about disease progression, with elevated stress levels significantly affecting QoL in diabetic patients with multimorbidity. Furthermore, comorbidities may restrict social engagement and occupational functioning. For instance, the coexistence of arthritis and diabetes can result in mobility limitations that hinder normal participation in social activities and employment. Such social isolation may in turn exacerbate psychological distress, forming a vicious cycle of multimorbidity, social withdrawal, and psychological imbalance, ultimately leading to further declines in QoL.

5.2.7. Self-care ability. The findings of this study indicate that better self-care ability is associated with better QoL, which is consistent with the results reported by Hysa et al. [44]. The reasons are as follows:

- (1) Patients with strong self-care ability are more likely to adhere to standardized behaviors, including dietary control, regular exercise, routine blood glucose monitoring, and medication compliance. Research has shown that self-care practices related to diet and exercise can significantly improve key health indicators such as glycated hemoglobin and reduce blood glucose variability [45]. Stable blood glucose levels substantially lower the risk of complications such as retinopathy and diabetic foot, thereby preventing associated physical dysfunction—forming the physiological basis for maintaining good QoL.
- (2) High self-care ability simplifies treatment processes and reduces additional burdens. Such patients are less likely to require frequent adjustments to their treatment plans due to proactive daily management, nor do they need to endure the constraints of complex treatments arising from disease fluctuations. For instance, studies grounded in Orem's self-care model have demonstrated that diabetic patients with good self-care ability experience a significantly lower incidence of complications, thereby avoiding the high medical costs and distress associated with such complications [46]. In contrast, poor self-care ability often leads to recurrent illness, which not only increases treatment complexity but also exacerbates physical and psychological suffering.
- (3) Studies have confirmed a significant positive correlation between self-care ability and QoL [47]. Patients with strong self-care ability can enhance their confidence in treatment by independently managing their condition, thereby reducing anxiety about disease progression. This positive mindset alleviates psychological stress. Moreover, a favorable physical condition enables them to participate normally in social activities and work. Research has also indicated that exercise-related self-care is associated with better QoL in the domain of social relationships, and active social engagement can further improve psychological well-being, creating a virtuous cycle that comprehensively enhances QoL [48].

5.3. Correlations among quality of life, self-regulatory fatigue, personal mastery, and health-promoting behaviors

5.3.1. Self-regulatory fatigue is associated with poorer quality of life. The findings of this study reveal a significant positive correlation between the self-regulatory fatigue and the quality of life in T2DM patients, consistent with the results

reported by Yuan [49] and Solberg et al. [50]. This indicates that higher levels of SRF are associated with lower QoL. The underlying reasons are multifaceted and interrelated, as elaborated below.

- (1) The negative association between SRF and QoL arises from the interplay of multiple pathophysiological and psychological processes—encompassing physiological mechanisms, cognitive patterns, behavioral responses, and social support—rather than a single causative factor. At its core, self-regulation functions as a form of passive compensation for the burden of disease rather than as effective disease management. The essence of SRF lies in the tension between subjective volition and objective physiological limitations, a conflict that may directly exacerbate diabetes-related pathological damage and, in turn, further intensify fatigue and diminish QoL. This mechanistic interpretation is supported by recent work from Solberg et al. [50], who demonstrated a strong negative association between SRF and health-related quality of life in individuals with T2DM.
- (2) Fatigue in these patients is often closely linked to glycemic fluctuations. When patients exert willpower to push through fatigue, the body's energy demands increase, while abnormal blood glucose levels exacerbate insufficient cellular energy supply, thereby intensifying fatigue and perpetuating a vicious cycle. This cascade may culminate in symptoms such as thirst, polyuria, and limb fatigue, directly impairing QoL in the physical function domain.
- (3) Fatigue may also serve as an early indicator of neuropathy or cardiovascular damage. If patients overlook these underlying pathological signals and rely on self-regulation to suppress discomfort, the risk of complications such as foot ulcers and angina pectoris may increase. Once such complications develop, they can substantially restrict patients' daily activities, leading to declines in both physiological and functional dimensions of QoL.
- (4) As a chronic condition, T2DM demands long-term self-management behaviors, including dietary control and blood glucose monitoring, which impose considerable psychological stress. Actively attempting to regulate fatigue adds an additional psychological burden. When fatigue persists and regulatory efforts prove ineffective, patients are prone to reduced self-efficacy, which may precipitate negative emotions such as anxiety and depression. Given that emotional state is a core dimension of QoL, anxiety and depression can directly lower patients' ratings of life satisfaction and mental health, thereby further compromising overall QoL.
- (5) Cognitive biases may intensify negative experiences. Some patients mistakenly interpret fatigue as a sign of severe illness and resort to self-regulation in an attempt to alleviate disease-related worries. However, such biases can heighten sensitivity to physical sensations, causing even normal fatigue to be perceived as disease progression and exacerbating anxiety. Concurrently, excessive rest may result in muscle weakness and poorer glycemic control, increasing susceptibility to fatigue and perpetuating a closed loop. Ultimately, patients become trapped in a cycle of concern over disease status and an inability to escape fatigue, resulting in sustained impairment of psychological well-being and subjective QoL.
- (6) Many patients lack professional knowledge regarding fatigue management, leading to misconceptions in their self-regulatory behaviors. For instance, some may attribute fatigue to inadequate nutrition and arbitrarily increase staple food intake, resulting in elevated blood glucose levels. Others may mistakenly believe that fatigue stems from insufficient physical activity and engage in excessive exercise even when glycemic control is unstable, increasing the risk of hypoglycemia or muscle injury. Such misguided regulatory strategies not only fail to alleviate fatigue but also disrupt overall disease management, diminishing patients' sense of control over their condition and further reducing QoL.

As the medical model continues to evolve, improving QoL in T2DM patients warrants increasing clinical attention. These findings suggest that healthcare providers should anticipate the potential for SRF in patients during long-term disease management, strengthen health education, and help patients develop an objective understanding of the physical and psychological challenges associated with the disease, thereby enhancing their capacity for self-regulation. Additionally, clinicians should engage in proactive communication with patients, offer empathetic support to address

reported psychological distress, assist in alleviating negative emotions, and guide patients toward a positive and optimistic approach to disease-related challenges. Concurrently, improvements in the medical resource support system are needed to bolster patients' confidence in disease management. Through these efforts, it may be possible to rebuild or preserve self-regulatory resources, reduce SRF levels, and ultimately enhance QoL.

5.3.2. Personal mastery is associated with better quality of life. This study revealed a statistically significant negative correlation between personal mastery and the quality of life score in T2DM patients, a finding that aligns with the results reported by Deng [51]. Given the inverse scoring of the DS-QoLS, this negative correlation correctly indicates that higher levels of personal mastery are associated with lower (i.e., better) QoL scores. In other words, patients with a stronger sense of control over their disease and life circumstances report a better quality of life. The underlying reasons are as follows: (1) Scientific and appropriate PM reflects a patient's effective autonomy in disease management, manifested as the ability to perform disease-related tasks independently and standardly—such as regular blood glucose monitoring, medication adherence, and adherence to a reasonable diet and exercise regimen. This directly reduces blood glucose fluctuations, lowers the risk of complications, and improves physical strength and comfort, thereby enhancing QoL in the physical function dimension. (2) When patients are able to manage their disease scientifically—for instance, by stabilizing blood glucose through self-management—they develop self-efficacy and experience reductions in anxiety and depression associated with poor disease control. Moreover, a clear understanding of the disease alleviates fear, promotes psychological and emotional well-being, and thus improves QoL in the psychological/spiritual dimension. (3) Scientific PM enables patients to proactively communicate with family members and healthcare providers, fostering greater family understanding and more precise medical guidance, which in turn strengthens social support. Additionally, stable disease status increases patients' willingness to engage in social activities, reducing disease-related isolation and improving QoL in the social relations dimension. (4) Patients with a scientific approach to PM are more likely to adhere actively to treatment, resulting in more stable therapeutic outcomes. They are also better able to view treatment adjustments rationally without becoming discouraged by minor fluctuations, contributing to a smoother treatment process and more favorable perceptions of the treatment experience, thereby enhancing QoL in the treatment dimension. Based on these findings, it is recommended that healthcare professionals support patients in developing the knowledge and skills needed for scientific disease management—for example, through tiered disease management training, simulated scenario exercises, and collaboratively established management goals—so as to strengthen PM and ultimately improve patients' QoL.

5.3.3. Health-promoting behaviors are associated with better quality of life. The results of this study demonstrate a significant negative correlation between health-promoting behaviors and the quality of life score in T2DM patients, a finding consistent with the study by Zhang [52]. Consistent with the scale's scoring, this negative correlation signifies that more frequent engagement in health-promoting behaviors is strongly associated with lower (i.e., superior) QoL scores. In essence, patients who more actively adhere to a healthy lifestyle experience a substantially better quality of life. The underlying reasons are as follows: (1) Adequate HPB—such as medication adherence, dietary regulation, and regular exercise—can directly mitigate blood glucose fluctuations, reduce the risk of complications including diabetic nephropathy and neuropathy, promote physiological improvements, and alleviate physical discomfort, thereby enhancing QoL. (2) Proactive engagement in healthy behaviors fosters a sense of disease control, alleviates feelings of helplessness and anxiety, and positively influences psychological well-being, which in turn reinforces daily activities and social participation, thereby improving self-efficacy. (3) Positive HPB can reduce the frequency of emergency department visits and hospitalizations, as well as medication-related expenses, thereby alleviating the economic burden and time demands associated with the disease, and indirectly enhancing patients' life satisfaction. (4) A high level of HPB contributes to improved health status and facilitates the recovery of social functioning, enabling patients to engage more actively in work and social activities, thus preventing disease-related social isolation and further strengthening QoL. These findings suggest that healthcare providers should develop individualized behavioral interventions and multidimensional

psychosocial support strategies, adopt a multidisciplinary collaborative approach, refine health education content, avoid theoretical indoctrination, and promote HPB among T2DM patients to ultimately improve their QoL.

5.4. Path analysis of quality of life, self-regulatory fatigue, personal mastery, and health-promoting behaviors

Based on stress coping theory and Roy's adaptation model, and informed by findings from the relevant literature, this study constructed a structural equation model incorporating four core variables in T2DM patients: SRF, PM, HPB, and QoL. The model fit indices met acceptable statistical standards, indicating a good model fit. The results revealed that SRF influenced QoL through four pathways:

- (1) In our hypothesized model, SRF showed a direct statistical pathway to QoL. This finding suggests that higher levels of SRF were associated with poorer performance of health behavior such as blood glucose monitoring, dietary control, and medication adherence, leading to increased glycemic variability and subsequent complications (e.g., neuropathy and infection), thereby directly reducing physical function. On the other hand, the exhaustion resulting from sustained self-imposed regulation contributed to anxiety and depression while undermining patients' confidence in managing their condition. These negative emotional experiences imposed physical and psychological burdens that ultimately diminished QoL. These findings suggest that healthcare providers should assess patients' levels of self-regulatory fatigue promptly and implement timely interventions to reduce SRF and improve QoL.
- (2) SRF influenced QoL through the mediating role of PM, with a mediation effect accounting for 28.7% of the total effect. This suggests that SRF affects patients' PM levels, which in turn affect QoL. According to the conservation of resources theory [53], the depletion of PM essentially reflects resource loss. Individuals' psychological and behavioral regulation depends on limited self-regulatory resources; when resource consumption outpaces replenishment, fatigue and maladaptation ensue. In patients with T2DM, SRF arises from the excessive consumption of self-regulatory resources due to prolonged self-management. Once these resources are exhausted, patients struggle to effectively manage health behaviors and develop a sense of losing control over their resources. This experience of exerting effort without maintaining resource balance directly undermines PM. As a core psychological resource for coping with illness, diminished PM further reduces patients' engagement in positive behaviors such as treatment adherence and social participation, contributing to poor glycemic control and social withdrawal, and ultimately impairing multiple dimensions of QoL.
- (3) SRF influenced QoL through the mediating role of HPB, with a mediation effect accounting for 44.6% of the total effect. This indicates that SRF affects QoL by shaping patients' HPB. The theory of planned behavior [54] posits that SRF undermines perceived behavioral control and behavioral execution. Prolonged volitional depletion leads patients to perceive an inability to sustain healthy behaviors; as perceived behavioral control declines, behavioral intention weakens, resulting in reduced HPB. Lower levels of health-promoting behaviors further diminish self-efficacy and outcome expectations, collectively reducing QoL through disease progression, heightened psychological distress, and restricted social functioning.
- (4) SRF also influenced QoL through the chain mediating effect of PM and HPB, with a mediation effect accounting for 11.3% of the total effect. The strong positive correlation between PM and HPB, which we disclosed, provides empirical support for this sequential pathway. However, it is critical to emphasize that the proposed ordering of PM → HPB is theoretically motivated (based on COR theory [53]), rather than empirically confirmed as a temporal sequence given our cross-sectional design. In this framework, PM represents a higher-order "meta-resource" (a global belief in one's ability to influence life events), while HPB are specific, effortful behavioral resources. The depletion of self-regulatory resources (SRF) first erodes the meta-resource of PM, and this erosion, in turn, leads to a decline in the specific behavioral resource of HPB. The significant correlation is a precondition for testing this hierarchical, causal

chain, and our SEM results suggest that PM acts as a partial gateway through which SRF impacts HPB, ultimately worsening QoL. According to the Conservation of Resources (COR) theory [53], individuals strive to protect their existing resources and acquire new ones. Personal mastery represents a key meta-resource—an individual's global belief in their ability to influence life events. Health-promoting behaviors are behavioral resources that depend on this meta-resource. Our proposed sequence posits that SRF, as a state of resource depletion, first erodes the higher-order meta-resource of PM. When patients lose confidence in their ability to control their disease (low PM), their motivation to engage in and sustain specific, effortful health behaviors (HPB) subsequently diminishes [54]. In contrast, a direct effect of SRF on HPB might represent a more immediate, temporary behavioral disengagement due to fatigue, while the chain-mediated path captures a more profound, cognitively mediated motivational deficit. This hierarchical depletion model provides a theoretical foundation for the observed chain mediation, where a decline in PM acts as a gateway to the deterioration of HPB, ultimately worsening QoL.

5.5. Limitations

While our study has several strengths, several limitations must be acknowledged.

- (1) Sampling bias and generalizability: Our use of convenience sampling from tertiary grade-A hospitals in two Chinese cities may have introduced significant selection bias. Patients treated in these high-level facilities likely have better access to specialized multidisciplinary diabetes care, structured education programs, and psychological support services compared to those managed in primary or rural healthcare settings. This differential access is particularly relevant for personal mastery. Specifically, patients in tertiary hospitals are more likely to receive empowerment-based education and individualized feedback, which directly build their confidence in controlling diabetes-related life events (i.e., PM). Consequently, the PM scores in our sample may be systematically higher than the true average for the broader T2DM population. This could lead to an underestimation of SRF levels and an overestimation of the protective effects of PM, HPB, and QoL. Therefore, our findings regarding “moderate” QoL and the observed effect sizes may represent a best-case scenario and might not be directly generalizable to underserved communities. Future multi-center studies should include a more diverse range of healthcare institutions and geographical regions to capture a wider spectrum of patients' psychological resources.
- (2) Gender imbalance: Our sample comprised 64.4% males, which, while consistent with the higher prevalence of T2DM in Chinese males, may limit the power for gender-specific analyses and the generalizability of findings to female patients. Given known gender differences in psychological and behavioral responses to chronic illness, future research should oversample female participants to formally test gender as a potential moderator in the mediation model.
- (3) Instrument validation: Although all scales demonstrated good internal consistency (Cronbach's α) in our sample, the Self-Regulatory Fatigue Scale (SRFS) was originally validated in young adults, and the Personal Mastery Scale (PMS) in migrant children. Their full psychometric properties, including construct validity via confirmatory factor analysis, have not been formally established in a middle-aged and older Chinese T2DM population. Potential measurement error from this source could affect the stability of the SEM results.
- (4) Unexplained variance in QoL: Our multiple linear regression model, while significant, explained only 41.8% of the variance in QoL (adjusted $R^2 = 0.418$). This indicates that over 58% of the variance remains unexplained. Other potent predictors, such as depression, diabetes distress, social support, and treatment satisfaction, were not measured in this study and should be integrated into future models to provide a more complete understanding of QoL in T2DM patients.
- (5) Causal inference and cross-sectional design: The most significant limitation of this study is its cross-sectional design. All data were collected at a single time point, which precludes any definitive conclusions about the direction of

causality or temporal sequencing among SRF, PM, HPB, and QoL. While our structural equation model tested theoretically derived pathways (based on the Conservation of Resources theory), the observed indirect effects are statistically equivalent to other possible causal models (e.g., lower QoL could increase SRF). Therefore, our findings should be interpreted as demonstrating statistical associations and indirect effects consistent with our hypotheses, not as evidence of causal mechanisms. The possibility of reverse or bidirectional relationships cannot be ruled out. For instance, a lower QoL might exacerbate self-regulatory fatigue, creating a negative feedback loop. Future research employing longitudinal cohort studies or randomized controlled trials is essential to establish temporal precedence and determine whether interventions targeting SRF can lead to improvements in PM, HPB, and ultimately QoL.

6. Conclusion

This study suggests that SRF is statistically associated with directly and indirectly poorer QoL in T2DM patients, primarily by being associated with lower PM and fewer HPB. These findings highlight the importance of addressing psychological and behavioral mechanisms in diabetes care. From a nursing perspective, interventions targeting SRF, strengthening PM, and promoting HPB may be beneficial.

Supporting information

S1 File. STROBE checklist.

(DOCX)

S2 File. Dataset used for the study.

(XLSX)

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